

purpose. The process of scrambling the message is encryption, and unscrambling it is decryption, performed through complex mathematical techniques.

Cryptography is the battlefield on which those trying to transmit information securely combat those attempting to decrypt or manipulate the information. More recently, cryptography has evolved to include applications like proving the ownership of information to a broader set of actors—such as public key cryptography—which is a large part of how cryptography is used within Bitcoin.

Encryption techniques have been employed for centuries. Julius Caesar used a simple method of encryption during times of war to inform his generals of his plans. He would send messages using letters that were three letters after the letter they were supposed to represent. For instance, instead of using the letters ABC in his message, he would write them as DEF and his generals would decrypt them to understand his intended message. Understandably, this form of encryption did not remain secure for long.³

A more recent example that was the subject of the movie *The Imitation Game* was the effort during World War II of a group of English cryptographers to decode the messages of Nazi Germany, which were encrypted by a coding device called the Enigma machine. Alan Turing, a luminary in machine learning and artificial intelligence, was a major player on the team whose efforts to break the